

Table 1: A comparison of the various deep learning frameworks

Index	TensorFlow	Caffe2	Theano	Keras	Torch	CNTK
1	Google created TensorFlow some time back primarily to replace Theano. Both these libraries are quite similar. It was said that Ian Goodfellow, who created Theano, also created TensorFlow. Google made it open source in 2015.	Caffe2 is the latest version of this model. This was created by Yangqing Jia who works at Facebook. This is the second most backed DL framework from Facebook after Torch/PyTorch.	Theano is a deep learning library framework developed by the Université de Montréal in 2007.	Keras is a Python based framework for deep learning and was developed by François Chollet, who works for Google.	PyTorch is an open source machine learning library for Python, based on Torch. It is developed by FB's AI research group-- Adam, Sam Gross, Soumith, and Gregory.	This computational toolkit was developed by Microsoft's research team during 2015-16.
2	The most used framework in the DL fraternity.	Generally preferred for RNN based models.	Cross-platform support.	Second most popular tool for DL.	Old ML based framework.	Most of the MS Windows based applications support this.
3	Developed using C/C++ backend engines. Performance is very good.	Developed using C++.	Performance is not that good for complex models.	Performance is good for most of the neural network layers.		Supports C++ and Python.
4	Supports both CPUs and GPUs across platforms.	Supports both CPUs and GPUs.	Supports both CPUs and GPUs.	Supports both CPUs and GPUs.		
5	Very good community support, and it provides good computation based features.	Backed by Facebook.		Many features are built in and it has good community support.		MIT licence and small community.
6	C++/ Python support.	Python/Java support.	Python support.	Python support.	Python support.	C++/Python support.

Traditionally, tensors have made less inroads into computer science. All these years computer science was more related to discrete mathematics and logical reasoning. But recent developments in machine learning have changed all that, and introduced the vector based mathematics and calculus of tensors. Here are a few basics:

- 1) Scalars are rank-0 tensors.
- 2) A Rank-1 tensor is a simple vector—all row vectors (1,2) and column vectors (2,1) are of these shapes.
- 3) A Rank-2 or 2D tensor is a simple 2X2 matrix. An example is: co-ordinates of a plane. A black and white image can be a 2D tensor.
- 4) Similarly, we have Rank-3 or 3D tensors and Rank-4 or 4D tensors, etc. A colour image of size (255,255, 3) is a 3D tensor (*Note: RGB is a three-colour channel*). A video

of a few minutes duration is an example of a 4D tensor, assuming that we have 60 frames per second, and RGB channels the size of (255, 255, 3, 3600).

 **Note:** The respective version of TensorFlow can be installed on your select choice of environment like Anaconda and Jupyter notebooks, Spyder, or any select framework that supports TensorFlow. The latest supported and stable version of TensorFlow is r1.13 or r1.12. TensorFlow 2.0 Alpha is also available now.

 **Note:** Make sure the respective Python version and matching TensorFlow version are both installed in your environment.